A Survey on Variable Reluctance Energy Harvesters in Low-Speed Rotating Applications

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***Abstract*— Energy harvesting converts ambient energy to electrical energy that can be used to power, for example, sensors and sensor systems. Variable reluctance (VR) energy harvesting is a suitable candidate for the conversion of rotary kinetic motion, an energy form commonly found in industrial applications. The implementation of a VR energy harvester (VREH), however, has a significant effect on its performance and is not well studied. In this paper, we therefore conduct a survey on different structures of VREHs. Six existing structures, previously used in VR sensors, are presented and analyzed according to their approaches for magnetic flux change improvement. Together with a newly proposed structure, these structures are evaluated based on a finite element analysis, and their results are compared. It is demonstrated that the choice of structure considerably affects the power output of the harvester and is dependent on the improvement approaches the structure utilizes. The newly proposed structure outperforms all existing structures with respect to power output and power density, which comes at a cost of higher parasitic torque generation. A 53-fold power improvement over the reference and an 1.2-fold power improvement over the next best structure is observed. As a result, applications of VR energy harvesting become viable even at low angular velocities.**

***Index Terms*— Energy harvesting, sensor systems, finite element analysis, electromagnetic induction, magnetic flux.**

I. Introduction

SENSORS and sensor systems play an increasingly important role in the operation, management and maintenance of industrial machinery [1]. A considerable number of these sensors are deeply embedded in the machine or added onto the machine, which complicates the provision of a dedicated infrastructure for such systems. Instead, they should, to a large extent, operate autonomously, which makes self-powered systems a desirable alternative.

Energy harvesting has in recent years become an alternative for the design of self-powered systems, aiming to reduce or eliminate the usage of battery-based sensor solutions, which, in turn, has a positive effect on maintenance costs and environmental pollution. Energy harvesting systems convert ambient energy sources, such as electromagnetic waves, thermal flows or kinetic movements, into electrical energy for the supply of low-power electronics [2].

Manuscript received January 16, 2018; accepted February 15, 2018. Date of publication February 21, 2018; date of current version March 22, 2018. This work was supported in part by the Knowledge Foundation through fund ASIS 20140323 and in part by the Vinnova through Grant 2017-03725. The associate editor coordinating the review of this paper and approving it for publication was Prof. Huang Chen Lee. *(Corresponding author: Sebastian Bader.)*

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Digital Object Identifier 10.1109/JSEN.2018.2808377

In most industrial environments, kinetic energy is commonplace in the form of vibrations, regular or random displacements, and driving forces. Rotational movements, in particular, are to be found in a majority of industrial machines, providing a ubiquitous and stable energy source to be exploited. Moreover, rotating parts are important to be monitored in order to feed control systems and to enable proactive maintenance, making rotational kinetic energy a suitable power source in close proximity to the sensor. Different electrical phenomena have been utilized to harvest the rotational energy, including, amongst others, electromagnetic induction, the Wiegand effect, and the piezoelectric effect [3].

In this article, we focus on energy harvesting from rotational kinetic energy based on the variable reluctance (VR) principle. Specifically, we investigate how different design decisions for variable reluctance energy harvesters (VREHs) affect their power output. The VR principle is based on Faraday's law of induction, where an electromotive force (EMF) is induced by a change in magnetic flux. In contrast to other electromagnetic transducers, in VREHs both the magnet and the pickup coil are stationary, which results in a robust design with low complexity. This principle has been extensively used in sensors, but its usage within energy harvesting is still very limited.

Previous research on variable reluctance as an energy harvesting method is presented in [3]–[6]. Häggström *et al.* [3] compare different harvesting methods in rotating applications, concluding that variable reluctance is one of the most promising techniques. Based on this, a basic VREH design has been investigated to estimate potential power levels. In [4], an autonomous wheel speed sensor is presented, which utilizes a commercial VR transducer to be self-powered. Similarly, in [5] and [6], variable reluctance energy harvesting is investigated as a method to power a sensor system for railroad applications. While these investigations demonstrate the feasibility of VREH in their respective applications, they are limited to basic VR designs and rather high rotational speeds ($>$300 rpm). A closer investigation on the effects of the harvesters' design and implementation is lacking.

In contrast, the VR principle has been used extensively in sensing applications. In this context it has been shown that the transducer design – particularly the shape and arrangement of the pickup unit – significantly contributes to its output signal [7]–[17]. This suggests that the pickup unit design for VREH is essential for the optimization of extracted output power, which is of particular importance at slow rotational speeds.

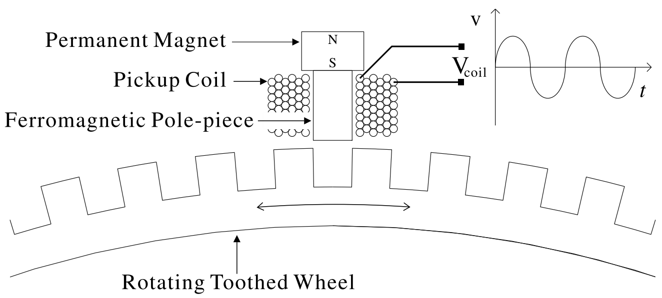


Fig. 1. A variable reluctance sensor outputs a time-varying voltage, the amplitude and frequency of which are related to the rotational speed and number of teeth of the rotating toothed wheel.

The overall objective of this work, therefore, is to provide a systematic study of different pickup unit designs for VREH, in order to support a better understanding of their influence on output power. This, in turn, will enable the optimization of VR transducers used in different energy harvesting applications, which will lead to a better understanding of VREH and thus improved transducer implementations. As such, we performed a survey on existing pickup unit structures concerning their effect on output power. The structures are first analyzed theoretically with respect to their capability to maximize magnetic flux changes in the pickup coil, which led us to propose yet another pickup unit structure. All structures are then compared based on a low rotational speed scenario in a 3D finite element analysis (FEA).

The results of this study show that moving from a basic VR structure to the best performing design yields a 53-fold output power improvement, whereas a simple and robust mechanical design is maintained. Although this improvement comes at a price of increased occupied space, even the power density is increased 30-fold.

II. Theory of Variable Reluctance Energy Harvesters

Figure 1 illustrates a simple structure utilizing the variable reluctance principle. It is composed of a rotating toothed wheel and a stationary pickup unit, including a permanent magnet (PM), a pole-piece and a pickup coil. The toothed wheel is typically mounted on an existing rotary target. An air gap exists between the pickup unit and toothed wheel, which is a necessary mechanical clearance for allowing the rotation of the toothed wheel.

The pole-piece and rotating toothed wheel are made from soft ferromagnetic materials, whereas the PM is made from hard ferromagnetic materials. The pickup coil is wound around the pole-piece, using it as its core.

A. Operation Principle

Working according to Faraday's law of electromagnetic induction, a VREH converts kinetic energy to electrical energy. An electromotive force (EMF) is induced across the pickup coil that couples to the time-varying magnetic flux caused by the rotating toothed wheel, and the variation in reluctance it results in. Connecting a load to the terminals of the coil generates a current, which can be used to power, for example, an autonomous sensor.

In contrast to other electromagnetic energy harvesters, the magnetic flux variation is not caused by a relative movement between coil and PM, but by the alternation of teeth and slots in the magnetic field lines, generated by the permanent magnet. The rotating wheel thus results in a time-varying flux, inducing an alternating voltage across the pickup coil, which can be described by

$$V_{\mathrm{c}}(t) = -\frac{d}{dt} \cdot \sum_{n=i}^{N_{\mathrm{c}}} \phi_i(t) \tag{1}$$

where V_{c} is the instantaneous voltage across the coil and ϕ_i is the flux linkage in any of the N_{c} turns of the pickup coil.

The frequency f_{c} (in Hz) of the coil voltage is proportional to the rotational speed ω (in rad s^{-1}) and the number of teeth of the rotating toothed wheel N_{t}, which is given by

$$f_{\mathrm{c}} = \omega \cdot N_{\mathrm{t}}. \tag{2}$$

In order to extract maximum power from such a device, the impedance of the load needs to be matched. At low frequencies (i.e., at low rpm) the coil inductance can be neglected and its resistance R_{c} is the dominating factor, such that maximum power is transfered with $R_{\mathrm{L}} = R_{\mathrm{c}}$. In that case, the load voltage can be expressed by

$$V_{\mathrm{L}}(t) = \frac{1}{2} \cdot V_{\mathrm{c}}(t). \tag{3}$$

Moreover, if the time-varying flux in (1) is an ideal sinusoidal wave and uniformly distributed in the coil, the peak-peak coil voltage under load is developed from (1) and expressed by

$$\begin{aligned} V_{\mathrm{L,pp}} &= -\frac{1}{2} \cdot N_{\mathrm{c}} \cdot \frac{\phi_{\max} - \phi_{\min}}{\frac{1}{2} \cdot T_{\mathrm{c}}} \\ &= -N_{\mathrm{c}} \cdot f_{\mathrm{c}} \cdot \Delta_\phi \end{aligned} \tag{4}$$

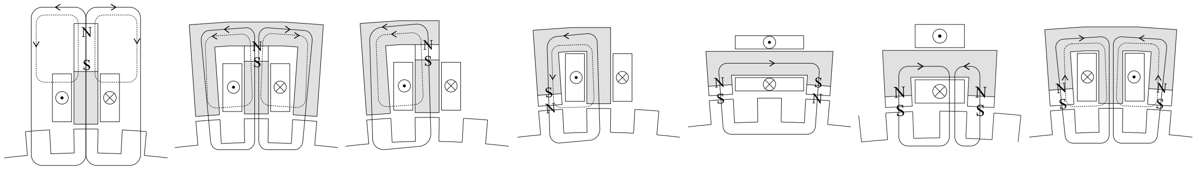
where Δ_ϕ is the difference between the maximum flux $\phi_{\max}$ and the minimum flux $\phi_{\min}$ in the coil, both of which are achieved at the aligned and unaligned position of tooth and pole-piece, respectively. $\frac{1}{2} \cdot T_{\mathrm{c}}$ is the time it takes for the rotating wheel to move between these two positions, that means the center of a tooth or slot facing the center of the pole-piece. Fig. 1 illustrates the pickup unit at the unaligned position, which is the pole-piece fully facing one of slots of the toothed wheel. Fig. 2 illustrates the two extreme positions for a simple VREH and exemplifies the resulting flux densities.

With the help of (2) and (4), the load power P_{L} can be expressed as

$$\begin{aligned} P_{\mathrm{L}} &= \left(\frac{\sqrt{2}}{4} \cdot V_{\mathrm{L,pp}}\right)^2 \cdot \frac{1}{R_{\mathrm{L}}} \\ &= \frac{(\omega \cdot N_{\mathrm{t}} \cdot N_{\mathrm{c}})^2}{8 R_{\mathrm{L}}} \cdot (\Delta_\phi)^2. \end{aligned} \tag{5}$$

This shows that the power output is proportional to the rotating speed ω, the number of teeth N_t, the number of coil turns N_c, as well as the magnetic flux difference Δ_ϕ between the two extreme positions; it is inversely proportional to the coil resistance (because $R_{\mathrm{L}} = R_{\mathrm{c}}$).

TABLE I
OVERVIEW OF COMPARED VR PICKUP UNIT STRUCTURES



Structure	ST-i	ST-mx	ST-nx	SS-nx	DSa-nx	DSr-nx	DSr-mx
Reference		[7]	[7]	[12]	[10]	[9]	
$\Delta\phi$ maximization methods	—	(a)	(a)	(a), (b)	(a), (b), (c)	(a), (b), (c)	(a), (b), (c)

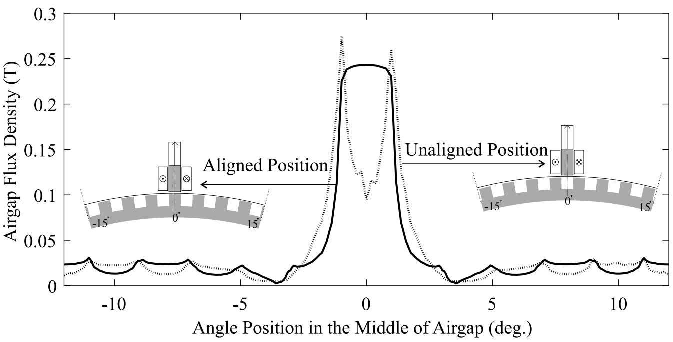


Fig. 2. Airgap flux density distribution of a simple VREH structure at aligned and unaligned positions. The total flux through the surface of the pickup unit shows a clear change, which will lead to a induced voltage.

B. Influences on Magnetic Flux Variation

According to (5), there are multiple factors that can improve power output. While several of these factors are typically defined by application constraints, the magnetic flux change Δ_ϕ can be affected by pickup unit design decisions. The main contributors to an increased Δ_ϕ are summarized in the following list:

(a) Implementing a soft-ferromagnetic return path to maximize main flux: This provides a main path for the flux, guiding the magnetic field. As a result, the magnetic field outside the provided path, which does not contribute to the flux variation inside the coil (leakage flux), is reduced.

(b) Placing both the PM and pickup coil close to the rotating toothed wheel: The pickup coil located in the proximity of the rotating toothed wheel is more directly coupled to the varying flux caused by the rotational interaction. PM location close to the rotating toothed wheel improves the flux variation in the coil, which has been shown in [18]–[20].

(c) Using Multiple PMs with a proper arrangement: In comparison to using a single PM that creates a single flux loop, the magnetic circuit of VREHs with dual PMs has an additional flux loop. Using the additional flux loop can increase the rate of change in flux, thus improving the power output. As flux difference has a quadratic effect, combined flux loops have a stronger influence as compared to alternating flux loops.

TABLE II
VREH STRUCTURE NOMENCLATURE

	Item	Definition
First part	ST	Single Top-mounted PM
	SS	Single Side-mounted PM
	DSa	Dual Side-mounted PMs (attracting conf.)
	DSr	Dual Side-mounted PMs (repelling conf.)
Second part	i, n or m	i-, n- or m-shaped pickup unit
	x	Span size

III. SURVEY OF VREH PICKUP UNIT STRUCTURES

In this section, seven pickup unit structures will be presented and analyzed with respect to their utilization of Δ_ϕ optimization methods, introduced in Section II-B. Six of these structures have previously been reported in the context of VR sensors, whereas the seventh is proposed by us based on the analysis presented herein. An overview of all structures is given in Table I.

In order to simplify the discussion of the different structures, a naming convention is helpful. As, to the best of our knowledge, a naming convention does not exist, we propose an own convention for the purpose of this article. A summary of this nomenclature is given in Table II. According to this, the name of each structure is composed of two parts, where the first part contains information about the number and position of PMs in the structure, and the second part acts as an overall descriptor of the structure itself. An ST-m2 structure, for example, contains a single (S) PM that is mounted on top (T) of the pickup coil, whereas the overall structure is formed like an m with a span size of two (2).

A. ST-i, ST-nx and ST-mx

ST-i, which has been presented in more detail in Section II, is the most basic structure of a VREH pickup unit. As the ferromagnetic pole-piece is limited to the core of the pickup coil, magnetic field lines return through a large air gap leading to great flux leakage. Most of this flux leakage will not contribute to a magnetic flux change in the pickup coil when the toothed wheel is rotating. As a result, the output signal (or output power in energy harvesting contexts) of this structure is low.

The structure ST-mx and ST-nx can be regarded as next step developments of ST-i. These structures add an external ferromagnetic member, which provides a main magnetic pathway and, thus, a return path for the field. ST-nx forms an overall n-shaped pickup unit, which means that a single return path is provided. ST-mx, is m-shaped and provides two parallel return paths. Both structures reduce flux leakage, resulting in a high flux variation in the surface spun by the coil, which leads to an increased output power, when the toothed wheel moves between aligned and unaligned position.

B. SS-nx

In addition to providing a ferromagnetic return path, SS-nx also utilizes the PM placement as an optimization factor for output power. Instead of residing on top of the coil, in SS-nx, the PM is moved to the end of the extended pole-piece and thus is placed directly adjacent to the air gap. This design maintains an overall n-shape with a single field loop, similar to ST-nx.

C. DSa-nx and DSr-nx

DSa-nx and DSr-nx are structures that utilize multiple magnets. Similarly to SS-nx, the PMs are placed adjacent to the air gaps. In DSa-nx, PMs are mounted in a supporting/attracting manner, resulting in a single field direction supported by both magnets. In contrast, DSr-nx places its PMs in non-supporting/repelling orientation, which forces the field lines of the two resulting loops to leave the ferromagnetic path. The exit location of the main field lines is changing with the rotation of the toothed wheel. Both DSa-nx and DSr-nx have an overall n-shaped structure with the coil wound around the pole-piece that connects both PMs, which enables a low profile. DSr-nx, however, is the only structure that does not spread a multiple number of teeth, but requires one PM to be aligned, while the other is unaligned.

D. DSr-mx

Finally, we propose DSr-mx, which is an extension of SS-nx resulting in two non-supporting PMs. As shown in Table I, this structure uses all three improvement methods, (a), (b) and (c), presented in Section II-B. DSr-mx is m-shaped with the pickup coil wound around the center pole (similar to ST-mx). As both magnets are placed in close proximity to the toothed wheel and also in a repelling manner, the two flux lines originating from each permanent magnet are forced through the center pole, resulting in a high flux variation in the core of the coil.

IV. Structural Parameters

The performance of all presented structures can be affected by altering physical dimensions. These physical dimensions include the span size of the pickup unit and the dimensions of the permanent magnet. As the effect of these alterations depends on the structure choice itself, an optimization needs to be performed on an individual basis. In order to allow for a fair comparison, these parameters are optimized for each structure prior to the comparison.

A. Span Size

Figure 3 defines the geometric parameters of a VREH exemplified on the structure SS-n2. The span size describes the physical width of the pickup unit in terms of the number of teeth it stretches. It is defined by the parameter α_{pitch}, which is a multiple of $\Delta\alpha$, with $\Delta\alpha$ being the angle difference between two adjacent teeth (or two adjacent slots) of the rotating wheel.

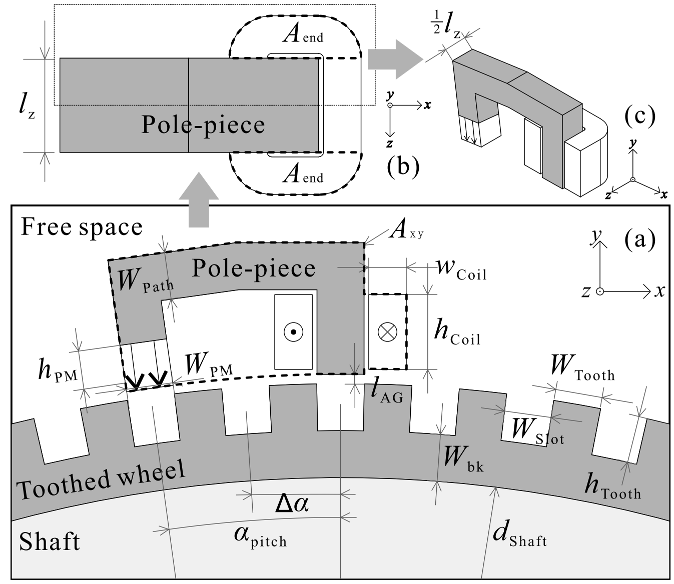


Fig. 3. Definitions of domains and geometric parameters in a VREH model SS-n2 in (a) cross-sectional view (xy-plane), (b) top view (xz-plane) and (c) perspective (view only showing the pickup unit). The permanent magnet is depicted as a rectangle with two arrows that indicate its magnetization direction. Two dashed contours, A_{xy} and A_{end}, indicate the areas occupied by the pickup unit in the cross-sectional view and by the coil in the top view, respectively.

In ST-nx and ST-mx, a smaller span size reduces the distance between PM and return path, which increases magnetic leakage (i.e., an increased magnetic field through the air). As a result, the magnetic field passing the toothed wheel is reduced, affecting the magnetic flux change in the pickup coil. A larger span size, in contrast, reduces this leakage at a cost of a overall larger physical size.

Similarly, SS-nx and DSr-mx experience a higher leakage if the span size is reduced. Although the placement of the PM adjacent to the air gap increases the coupling to the toothed wheel, leakage through the air between different parts of the pole-piece remains and is related to their distance (i.e., the span size). Even for DSa-nx and DSr-nx influences due to span size variation were observed.

As the influence of span size on output power was observed to have a different magnitude for the different structures, span size needs to be taken into account in order to result in a fair structure comparison.

B. Permanent Magnet Dimension

The energy product of a PM in a magnetic circuit with a given air gap size is a known design parameter in a number of electromagnetic applications. As such, it is also suitable to be considered for VREH optimization. In order to achieve maximum power output for a given VREH structure, the PM's energy product needs to be maximized. Maintaining the same magnet material and circuit configuration, the energy product of the PM can be affected by its physical dimensions,

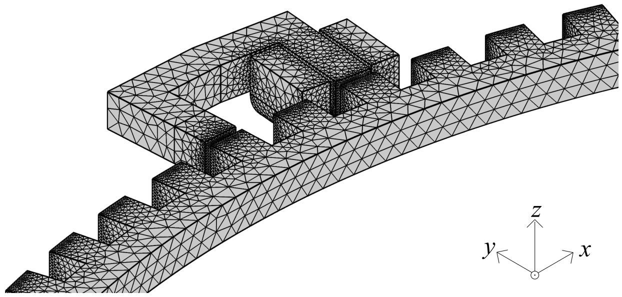


Fig. 4. Mesh configuration of structure SS-n2 (without air domain). The completed meshing consists of 1146923 domain elements, 132179 boundary elements, and 5687 edge elements.

TABLE III
PARAMETERS OF THE VREH MODEL

Parameter	Definition	Value
ω	Angular velocity	$1.0576\,\mathrm{rad\,s^{-1}}$
N_{Tooth}	Wheel teeth number	90
f_{Coil}	Frequency of coil voltage	$15.15\,\mathrm{Hz}$
T_{Coil}	One cycle of f_{Coil}	$1/f_{\mathrm{Coil}} = 0.066\,\mathrm{s}$
t_0	Initial time	$0\,\mathrm{s}$
t_{stop}	Stop time	$0.066\,\mathrm{s} = T_{\mathrm{Coil}}$
Δ_t	Step size	$0.001\,\mathrm{s}$
N	Number of steps	$65 = \frac{t_{\mathrm{stop}} - t_0}{\Delta_t} - 1$
l_z	VREH length in the axial direction (z-direction)	$10\,\mathrm{mm}$

especially by the dimension co-linear to the magnetization direction [21].

The leakage flux and the fringing effect, however, are other factors that define the optimal dimension of the PM, making a purely analytical estimation of optimal PM size difficult. Therefore, a parametric sweep of the PM's dimension is performed for each of the structures. In order to simplify this optimization process, the PM volume is only changed by sweeping its height, the dimension co-linear with the magnetization direction, whereas the PM width and length are maintained at the same dimensions as the pole-piece.

V. Simulation Setup

The comparison of the different VREH structures is performed based on a finite element analysis, conducted in COMSOL Multiphysics. All structures are compared in the same scenario with a rotating shaft of 270 mm diameter and a rotational speed of approximately 10 rpm. A more detailed description of the configuration of this FEA is given below.

A. FEA Configuration

The VREH analysis in COMSOL Multiphysics is performed as a time-stepping, three-dimensional FEA. A VREH geometric model is depicted in Fig. 3, exemplified on a SS-n2 structure. The structural rotation is modeled using the *Rotating Machinery, Magnetic (rmm)* interface in COMSOL. The rotating part of the model includes the toothed wheel, the shaft, as well as a part of the air gap between toothed wheel and pickup unit. Rotation occurs at a constant angular velocity $\omega = 1.0576$ rad s^{-1} ($\approx$10.10 rpm).

In order to reduce the model size and computation time, the full geometry shown in Fig. 3(b), an axial mirror symmetry model, is halved along the xy-plane symmetry plane shown in Fig. 3(c). Thus, the depth of the simulated model is reduced to $\frac{1}{2} \cdot l_z$ (= 5 mm). However, the coil voltage and current of the full model are computed automatically by setting the coil length multiplication factor in the *rmm* interface to $F_L = 2$.

Fig. 4 exemplifies the corresponding meshing configuration on the SS-n2 structure. A fine meshing in the air gap and the core of the coil are of particular importance, whereas meshing in the rest of the structure can be slightly coarser.

The VREH power output is estimated by connecting a resistive load to the pickup coil, utilizing the *Electrical Circuit (cir)* interface in COMSOL. As given by (2), at the low frequency, the coil inductance can be ignored, and its resistance R_{Coil} is the dominating factor of the coil impedance. In order to achieve maximum power transfer, the load resistance thus is selected to be equal to the pickup coil resistance.

Due to its time-dependency, output power is calculated as RMS power, being the average of instantaneous power over the time duration $t_0 \sim t_{\mathrm{stop}}$ $(= T_{\mathrm{Coil}})$, which is expressed by

$$P_{\mathrm{RMS}} = \frac{1}{t_{\mathrm{stop}} - t_0} \cdot \frac{\Delta_t}{2N} \cdot \sum_{n=0}^{N} \left[P(t_n) + P(t_{n+1}) \right] \qquad (6)$$

where t_0, t_{stop}, Δ_t and N are given in Table III, and $P(t_n)$ is the instantaneous power of the load R_{Load} at a certain time t_n, which is $P(t_n) = V^2(t_n)/R_{\mathrm{Load}}$.

B. Domain Configuration

Different domains in the model are assigned specific material properties to reflect the different materials used in the real structural configuration. This includes the magnetic properties of the PM and magnetic steel properties of the pole-piece and toothed wheel. A more detailed description of material property assignment in different domains of the model are given in Table IV. Temperature effects are not considered in this comparison and hence all material properties are modeled to be temperature invariant with material properties for 20 °C.

The magnetization direction of the PM is co-linear with its height vector. The PM operating flux density B_{PM} hence depends on the orientation of the height vector in the xy-plane and is modeled by COMSOL Multiphysics as

$$\begin{bmatrix} B_{\mathrm{PM(x)}} \\ B_{\mathrm{PM(y)}} \\ B_{\mathrm{PM(z)}} \end{bmatrix} = \mu \cdot \begin{bmatrix} H_{\mathrm{PM(x)}} \\ H_{\mathrm{PM(y)}} \\ H_{\mathrm{PM(z)}} \end{bmatrix} + B_r \cdot \begin{bmatrix} \sin(\alpha_{\mathrm{pitch}}) \\ \cos(\alpha_{\mathrm{pitch}}) \\ 0 \end{bmatrix} \qquad (7)$$

where μ is the product of vacuum permeability (μ_0) and the relative permeability of the PM material (μ_{r}), B_r is the permanent flux density, and α_{pitch} is the pitch angle shown in Fig. 3. For example, the pitch angle of the SS-n2 is $\alpha_{\mathrm{pitch}} = 2 \cdot \Delta\alpha = 2 \cdot 2\pi/N_{\mathrm{Tooth}} = 2\pi/45$ $(= 8^\circ)$.

The coil dimensions for most structures are the same and do not change with span size variations. An exception to this are DSa-nx and DSr-nx structures, where coil length is proportional to span size as exemplified in Fig. 5. Due to the horizontal orientation of the coil in these structures, it is

TABLE IV
OVERVIEW OF DOMAIN CONFIGURATIONS

Domain	Parameter	Definition	Material selection and/or modeling specification
Permanent magnet	h_{PM} $W_{\mathrm{PM}} = 5\,\mathrm{mm}$	PM height PM width	NdFeB-N50. $\mu_{\mathrm{r(PM)}} = 1.05$, $B_r = 1.40\,\mathrm{T}$, h_{PM} is swept in the range of $1{\sim}10\,\mathrm{mm}$ (stepsize $0.1\,\mathrm{mm}$).
Pickup coil	$\mathcal{V}_{\mathrm{Coil}} = 1823.08\,\mathrm{mm}^3$ $R_{\mathrm{Coil}} = 61.62\,\Omega$	Coil volume Coil DC resistance	A multi-turn coil consists of 1306 round copper wires with the diameter of $0.143\,\mathrm{mm}$ and the insulation of $10.5\,\mu\mathrm{m}$ (AWG 35 single build magnet wire) in a square pattern arrangement [22].
Pole-piece	$W_{\mathrm{Path}} = 5\,\mathrm{mm}$	Pole-piece width	M235-35A [23]. The modified pole-piece forming the return path presents varies shapes but has a fixed width.
Coil bobbin	$t_{\mathrm{Bob}} = 0.5\,\mathrm{mm}$	Coil bobbin thickness	Air (COMSOL material library). A clearance between the pickup coil and the pole-piece is used to represent the coil bobbin.
Toothed wheel	$W_{\mathrm{Tooth}} = h_{\mathrm{Tooth}} = 5\,\mathrm{mm}$ $W_{\mathrm{Slot}} = 5\,\mathrm{mm}$ $W_{\mathrm{bk}} = 5\,\mathrm{mm}$	Single tooth width and height Slot width Back iron width	M235-35A [23]
Air gap	$l_{\mathrm{AG}} = 1\,\mathrm{mm}$	Air gap size	Air (COMSOL material library). The clearance is between the pickup unit and the toothed wheel.
Shaft	$d_{\mathrm{Shaft}} = 270\,\mathrm{mm}$	Shaft diameter	Air (COMSOL material library)
Free space	$r_{\mathrm{Free}} = 175\,\mathrm{mm}$ $l_{\mathrm{Free}} = 25\,\mathrm{mm}$	Radius of cylinder Length of cylinder	Air (COMSOL material library). A cylindrical free space centered on z-axis

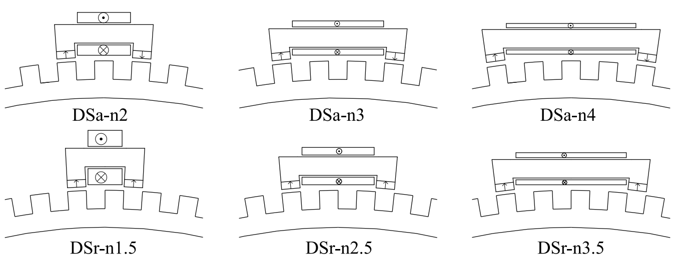


Fig. 5. In both DSa-nx and DSr-nx the coil is placed horizontally, which makes adjustments in length along with span size variation reasonable. As coil volume (number of turns) is maintained, longer span sizes lead to longer, but thinner coils.

unrealistic to assume static coil lengths even if span size increases. Therefore, coil size variation for these structures was perceived to be the most fair solution. However, the number of turns and thus the volume of the coil is kept the same even for these structures.

C. Parameter Sweeps

As described in Section IV, the span size and PM height can have a considerable effect on output power, but the extent of their influence is structure dependent. In order to lead to a fair comparison, therefore, multiple configurations of the same structure are evaluated. With respect to span size, three different configurations for each structure are investigated. For each configuration, the optimal PM size is determined by performing a sweep of the PM height in 0.1 mm steps. Only the results for optimal PM configurations are reported.

D. Power Density

Besides power output, power density is an essential parameter for the comparison of energy harvesting devices. Power density, herein, is estimated as

$$D_p = \frac{P_{\mathrm{RMS}}}{\mathcal{V}_{\mathrm{pickup}}} = \frac{P_{\mathrm{RMS}}}{2 \cdot h_{\mathrm{coil}} \cdot A_{\mathrm{end}} + l_z \cdot A_{\mathrm{xy}}} \tag{8}$$

where $\mathcal{V}_{\mathrm{pickup}}$ is the volume of the pickup unit, composed of the pole-piece, permanent magnet and coil. An example of these areas for SS-n2 are marked by the dashed contour in Fig. 3.

E. Torque Ripple

Finally, adding a VREH transducer to a rotating, metallic shaft introduces a torque on the shaft, which will influence its rotation. This torque is mostly undesired and has two contributing components, namely cogging torque and electromagnetic torque. The cogging torque is caused by the permanent magnets in the structure and is primarily dependent on its total volume and location, as well as the geometry of PM, pole-piece and toothed wheel. Cogging torque can lead to mechanical vibrations and acoustic noise [24]. The electromagnetic torque is caused by the electromagnetic force generated by the induced current in the pickup coil. As such, it is proportional to the extracted power of the VREH.

The torque ripple of each VREH structure is estimated according to Arkkio's method [25], which is expressed as

$$\tau = \frac{2}{\mu_0 \cdot l_{\mathrm{AG}}} \cdot \int_0^{l_{\mathrm{Free}}} \int_S B_r \cdot B_\varphi \cdot r_{\mathrm{AG}} \, \mathrm{d}S \, \mathrm{d}z \tag{9}$$

Herein, l_{Free} is the depth of the simulated free space domain, l_{AG} is the air gap size, S is the surface along the air gap (in the xy-plane), r_{AG} is the air gap radius (i.e., the distance from the center of the shaft to the the center of the air gap), and B_r, B_φ are the radial and tangential components of the flux density in S, respectively. The factor of two compensates for the fact that only half of the symmetric model is simulated.

VI. RESULTS AND DISCUSSION

In this section, we present and analyze the results obtained from the 3D FEA. Simulations are performed on each structure with different span sizes to predict output voltage, current,

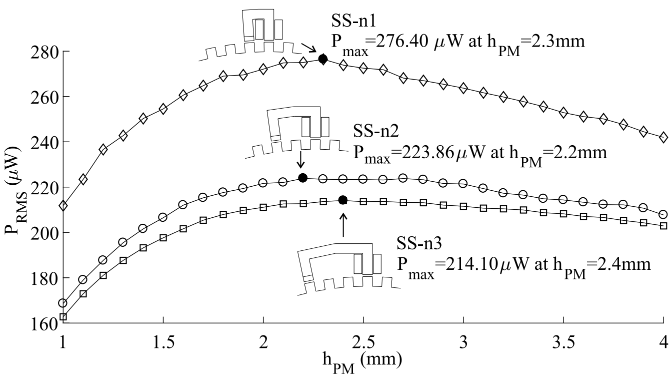


Fig. 6. Power output of SS-nx structures at different PM heights. The sweep of PM heights allows for the identification of optimum PM height for the given structures (marked black).

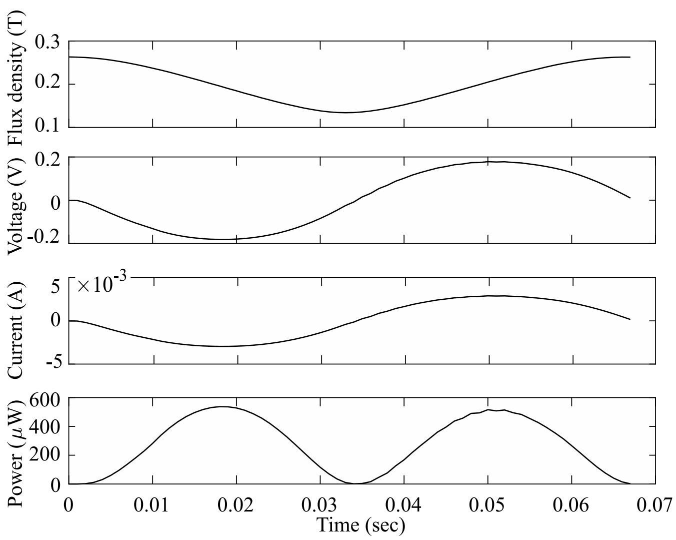


Fig. 7. Illustration of air gap flux density B, coil voltage V_{L}, current I and power output P, exemplified for SS-n1 with a PM height of 2.3 mm over one cycle (T_{Coil}).

TABLE V
RESULTS FOR VREHs WITH SINGLE PM

Structure	Overall span	Power increase [× ST-i]	Space increase [× ST-i]	Power Density [nW/mm^3]	h_{PM} [mm]
ST-n1	1	4.90	1.52	16.76	3.3
ST-n2	2	5.17	2.42	11.09	5.5
ST-n3	3	5.18	3.29	8.16	6.1
ST-m2	2	4.99	2.02	12.80	2.9
ST-m4	4	5.40	3.47	8.07	3.4
ST-m6	6	5.39	6.66	4.19	4.3
SS-n1	1	17.56	1.34	68.11	2.3
SS-n2	2	14.22	1.91	38.62	2.2
SS-n3	3	13.60	2.49	28.27	2.4
ST-i	0	15.74 μW	3036.17 mm^3	5.18	10

TABLE VI
RESULTS FOR VREHs WITH DUAL PMs

Structure	Overall Span	Power increase [× ST-i]	Space increase [× ST-i]	Power Density [nW/mm^3]	h_{PM} [mm]
DSa-n2	2	38.94	1.29	156.18	1.2
DSa-n3	3	42.27	1.47	149.42	1.1
DSa-n4	4	44.71	1.74	133.42	1.2
DSr-n1.5	1.5	21.41	1.25	88.87	1.3
DSr-n2.5	2.5	28.17	1.31	111.15	1.3
DSr-n3.5	3.5	32.05	1.54	108.00	1.3
DSr-m2	2	53.40	1.76	157.66	3.3
DSr-m4	4	42.17	2.90	75.26	4.6
DSr-m6	6	37.58	4.08	47.71	5.4
ST-i	0	15.74 μW	3036.17 mm^3	5.18	10

and power. For each configuration, the optimum PM height is determined through simulation sweeps. The individual results of these steps are exemplified for the structure SS-nx in Figs. 6 and 7. As a result, Tables V and VI present the span size, power generation, occupied space, power density, and PM height of the individual structures that are used for comparison, separated by the number of PMs in the pickup unit. Moreover, Fig. 8 graphically compares the configurations with the highest power density in their respective group. Generally, ST-i is used as the reference, and result performances are expressed as relative deviations from the performance of ST-i. Finally, Fig. 9 shows the torque ripple introduced by the VREH structures to the rotating shaft.

A. Power Output and Power Density

The maximum output power obtained from ST-i is 15.74 μW at a PM height of 10 mm. As such, this value will be used as a reference value for any further performance comparisons. As was previously presented, a low power output of this structure was expected.

Providing a ferromagnetic return path, ST-nx and ST-mx obtain a higher output power than ST-i. Optimizing the span-size and PM height, power output improvements by a factor of 4.90 and 4.99 can be achieved for ST-nx and ST-mx, respectively. However, the highest power output for these structures comes at a cost of increased volume, which affects their power density. The best power density is achieved with the smallest span-size, reducing power output improvements to a factor of approximately 5 for both structures. Although ST-mx has a lower reluctance than ST-nx and thus achieves higher output powers, its additional space occupation leads to a considerably lower power density.

In comparison to ST-nx, SS-nx demonstrates a considerable improvement in both power output and power density due to its PM position. With an 17.56-fold power improvement over ST-i, the PM location is an important factor for achieving high power output in VREHs. Span-size increment, moreover, does not improve power output for this structure, leading to SS-n1 providing the highest power output and power density.

DSa-nx and DSr-nx can provide a higher power output than single-PM structures. Both of them have increased output power with increasing span-size, but DSa-nx achieves higher power output and higher power density than DSr-nx in all tested cases. DSa-nx also requires the smallest PM volume to achieve its optimal performance.

Finally, DSr-mx provides the highest power output and power density of all tested structures. With a power output of over 840μW in the presented test case, it provides a 53-fold

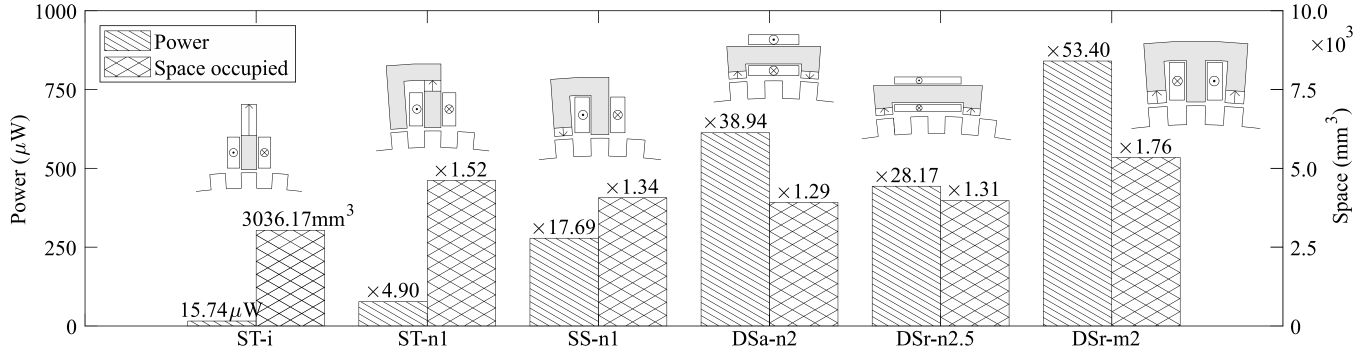


Fig. 8. Comparison of the power generation and space usage for the structure configuration with the highest power density. Two rectangles with dot and cross present the pickup coil. The rectangle with a single arrow presents the permanent magnet with magnetization direction. Numbers given in the figure indicate relative increments over the reference ST-i.

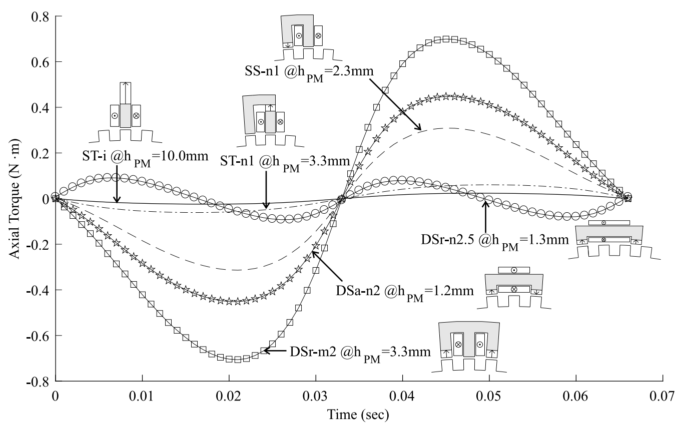


Fig. 9. Torque ripple introduced by the different VREH structures to the rotating shaft over one cycle (T_{Coil}). A generally increased torque with higher output power and larger PM size is observed.

improvement over the reference structure ST-i. Although DSr-mx has a comparatively large space requirement, its power density is with approximately 158 nW mm^{-3} the highest of all structures evaluated.

B. Structure Comparison

The results presented in Tables V and VI, as well as Figs. 8 and 9 clearly show that the pickup structure choice and its configuration has a significant impact on the achievable power, power density and generated torque of a VREH. Each approach for increasing flux variation, presented in Section II, will increase the power output of the structure utilizing this approach. As a result, it can be seen that structures incorporating more of these approaches than others also demonstrate a higher performance with respect to their power output.

Some of these approaches, however, require a larger space occupation (e.g., adding a ferromagnetic return path). As such, power density is reduced, which may lead to drawbacks in applications where space, and thus power density, is of importance. In most of the presented cases, however, the gained power output outweighs the additional space occupation, leading to an overall more power dense solution.

Moreover, power output, as well as the PM volume and location, determine the amount of torque that the structure will generate on the rotating shaft. Fig. 9, shows that a higher output power typically comes with an increased torque level, which may affect system efficiency. Whether this torque level is a problem, however, is application-specific and needs to be investigated on an individual basis.

Judging by power output and power density, the structure DSr-m2 has the highest performance of all evaluated structures. An output power increase by a factor of about 53 with respect to the reference structure (ST-i) is observed. In comparison to the next best structure (DSa-n4), DSr-m2 shows an improvement in output power and power density of approximately 20 percent.

Nevertheless, power output and power density might not be the only concerns in the application of VREHs. The n-shaped structures generally have a smaller space requirement in comparison to m-shaped structures, and particularly DSa-nx and DSr-nx structures can be implemented with a low profile. Due to their small PM volume, these structures also demonstrate a considerably lower torque in comparison to DSr-m2. This can lead to lower levels of mechanical vibrations and acoustic noise, and may be beneficial to applications sensitive to overall efficiency.

C. Limitations

All results are obtained under the setup presented in Section V. This setup has the following limitations. (i) Structures are only evaluated in one application scenario. This scenario has a fixed shaft diameter, angular velocity and tooth wheel configuration. (ii) Not all dimensions are altered to optimize each individual pickup unit structure. PM volume, for example, is only optimized by adjusting its height, whereas its width and depth remains constant. Similarly, the width and depth of the pole-piece is unaltered. Coil dimensions and number of turns are unaltered.

While these limitations simplify the analysis and some deviations from absolute results in experimental measurements are to be expected, none of the presented structures is given a particular advantage nor disadvantage. As such, relative differences in the performances of the different pickup unit structures are expected to be accurate.

VII. Conclusion

In this paper, seven different pickup unit structures have been presented and were compared with respect to their performance in variable reluctance energy harvesting. FEA results demonstrate that there is a significant performance difference in power output and power density for the evaluated structures. Three key approaches to the performance improvements have been identified, which are (i) the existence of a ferromagnetic return path, guiding the flux through the surface spun by the pickup coil; (ii) the placement of the permanent magnet(s) in the magnetic circuit, leading to a stronger coupling with the toothed wheel when placed adjacent to the air gap; and (iii) the utilization of multiple PMs, leading to a larger flux change in the coil.

Comparison results furthermore show that the performance of a structure increases with the number of improvement approaches it utilizes. As a result, all dual-PM structures perform better than single-PM structures. Based on the analysis of improvement approaches presented, a new structure (DSr-mx) has been introduced as a natural development of SS-nx and ST-mx. This structure achieved the highest power output and power density of all structures evaluated. In comparison to the reference structure (ST-i) a 53-fold improvement in power output could be observed, which is a 20 percent increase over the second best performing structure. Moreover, these improvements come at low cost, maintaining a simple and robust mechanical structure.

High power output and high power density are important metrics for VREHs, particularly in rotation applications with low angular velocity. Electronic systems (e.g., wireless sensors) can thus be reliably powered by the VREH even at low rotational speeds. For example, the FEA estimated the power output of the DSr-m2 structure to be approximately 840 μW at a rotational speed of about 10 rpm (270 mm shaft diameter), a power level sufficient for many duty-cycled, low-power wireless sensor systems.

Moreover, this survey provides a guideline for the selection of the right type of VREH structure, based on application requirements such as power output, torque ripple, available space, form factor and cost.

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